

JONATHAN ONG CHONG AN

jonongca@gmail.com | +65 89205102 | jonongca.com
linkedin.com/in/jonongca | github.com/JonOng2002

Availability: January 2027 – August 2027

Singapore, SG

EDUCATION

Bachelor of Science, Information Systems (Product Development)

Jan 2023 – Dec 2027

Singapore Management University | Second Major: Computer Science (Artificial Intelligence)

SUMMARY

Information Systems undergraduate with experience in large-scale data pipelines, backend systems, cloud infrastructure, and applied AI. Skilled in Python, SQL, PySpark, ETL, data validation, and production reliability, with hands-on experience building enterprise data workflows and AI-enabled systems.

EXPERIENCE

AI & Data Engineering Intern, Inland Revenue Authority of Singapore (IRAS)

May 2026 – Nov 2026

- Migrated 6 legacy SAS ETL scripts to PySpark for monthly batch-processing workflows handling multi-million-row tax datasets, supporting downstream analytics and reporting.
- Contributing to the conversion of an on-premises Python data ingestion pipeline to PySpark for cloud deployment, reviewing data flows and transformation logic to improve scalability, maintainability, and production readiness.
- Developed reusable validation checks across schemas, joins, timestamps, and business rules to identify data discrepancies and improve confidence in migrated pipeline outputs.
- Converted two form data processing pipelines from batch Python scripts into PySpark notebooks, enabling business users and data analysts to continue data processing during migration while proposing standardised CSV ingestion and ABFSS path configuration for the data engineering team.
- Integrated internal search APIs into an enterprise assistant's RAG retrieval workflow, improving access to trusted organisational knowledge and supporting more relevant, evidence-grounded responses.
- Supported the migration of internal machine-learning APIs to Azure ML by standardising model artefact management, strengthening logging and exception handling, and introducing automated pytest checks that increased the CI test success rate by approximately 50%.

DevOps Engineer Intern, YTL PowerSeraya

Jan 2026 – Apr 2026

- Built and rolled out Azure DevOps multi-stage YAML pipelines for 4 internal gas-management applications, reducing deployment time from 60 to 20 minutes and improving release consistency.
- Designed a self-hosted agent approach for secure IaaS deployments within an isolated VNet, enabling CI/CD execution on application VMs and documenting the workflow for developer adoption.

Cloud Engineer Intern, Global Enterprise Mobility (GEM)

May 2025 – Aug 2025

- Built Jenkins CI/CD pipelines using Terraform-provisioned AWS infrastructure and GCP Secret Manager, reducing release time from 40 to 20 minutes while improving deployment repeatability and credential security.
- Deduplicated webhook events using a DynamoDB idempotency layer, resolving a production defect that affects approximately 12–15% of transactions.

PROJECTS

PitchPerfect — AI Evaluation Platform | Product Owner & Data Pipeline Lead

Jun 2026 – Present

- Product Owner in a six-member Agile Scrum team developing an AI evaluation platform for the Lee Kuan Yew Global Business Plan Competition, reducing the time it takes to evaluate pitch decks by 80%.
- Own the ingestion and retrieval workstream, translating a manual pitch-deck review process into structured feature extraction, evidence retrieval, and readiness-assessment stages.
- Design a pipeline to ingest labeled pitch decks, extract rubric-aligned features and metadata, generate embeddings, and retrieve supporting evidence for consistent evaluation.

YilongMa — Retrieval-Augmented Generation Data Pipeline

Aug 2025 – Dec 2025

- Built an end-to-end data and retrieval pipeline using Python, ChromaDB, Llama 3.2 1B, and hybrid search to generate grounded responses from a transcript-based knowledge base.
- Processed more than 50 hours of audio using Whisper and developed evaluation notebooks to assess retrieval relevance and response grounding.

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google IT Automation with Python | Java Certified Foundations Associate

SKILLS

Languages & Frameworks: Python, SQL, Java, JavaScript, Bash, FastAPI, Flask, Vue.js, React.js, Pytest, Git

Data Engineering: PySpark, Pandas, Databricks, Azure Synapse, ETL, Hive, SAS, Azure Data Lake Storage

Cloud & Infrastructure: AWS, Azure, Google Cloud Platform, Docker, Terraform, Jenkins, CI/CD

AI/ML: RAG, Embeddings, Hybrid Retrieval, LLM Evaluation, Prompt Engineering, Agent Harness